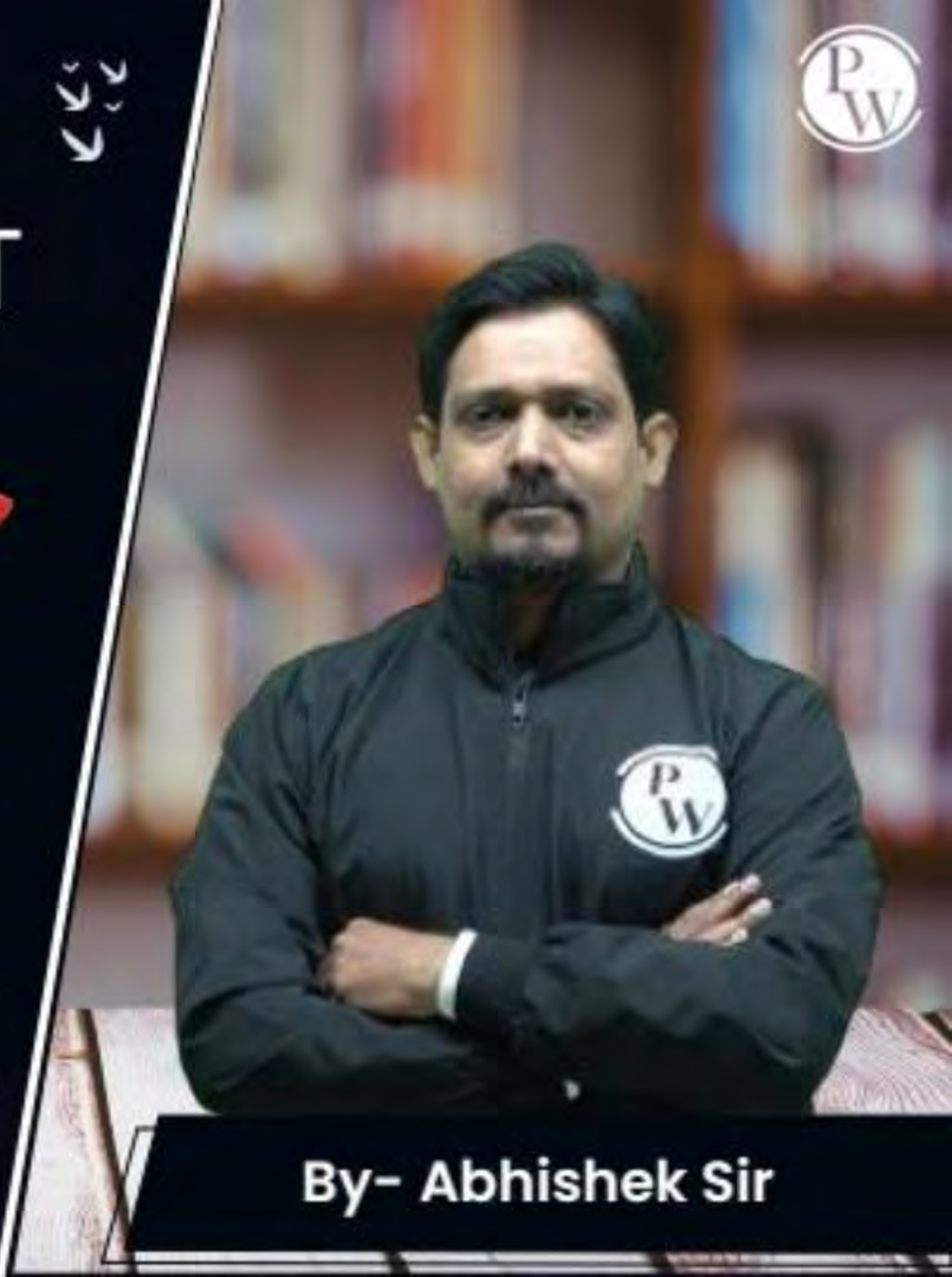
Computer Science & IT



Fundamental of C Language

Introduction

Lecture No. 01



By- Abhishek Sir

Topics to be Covered



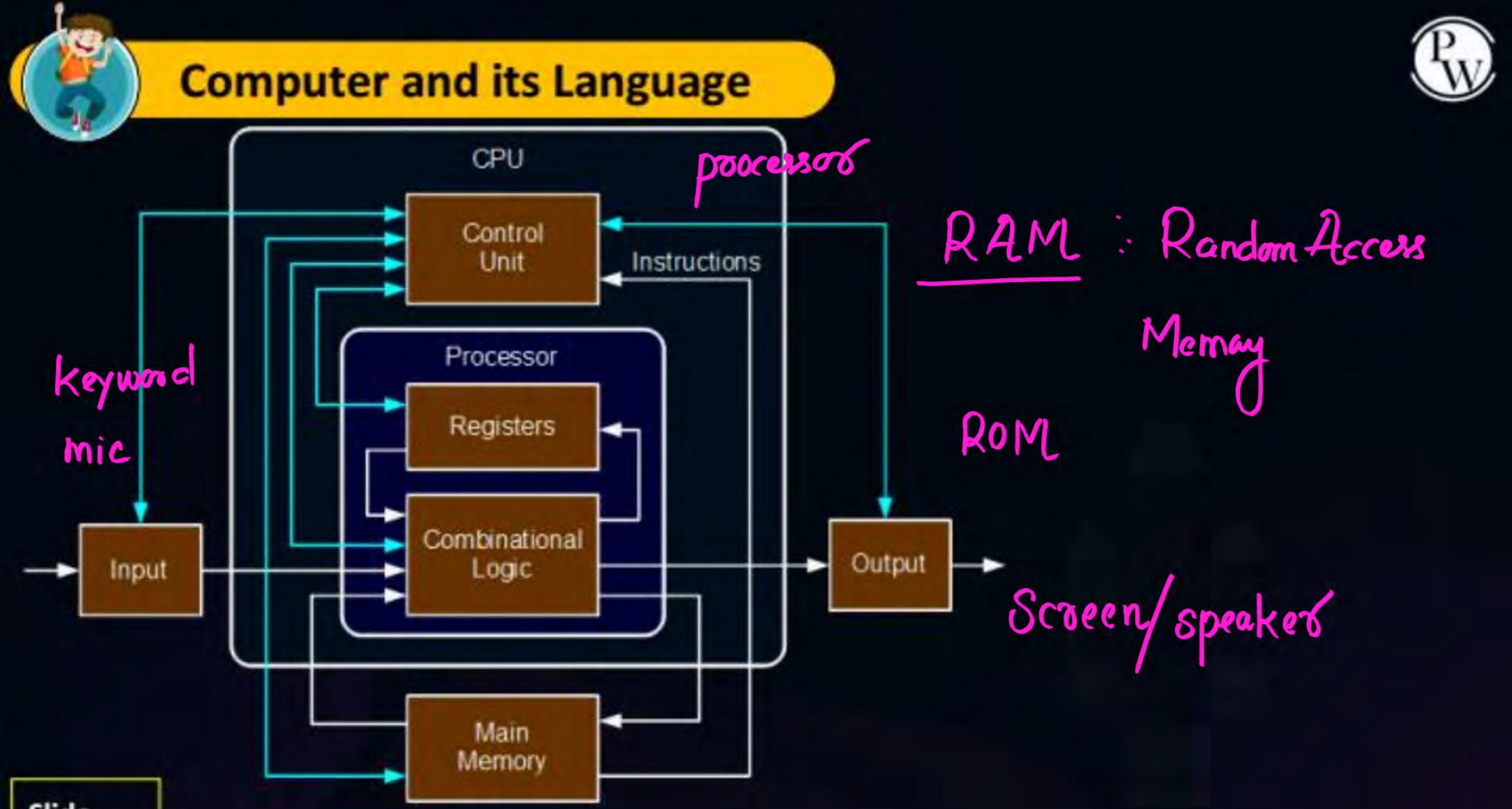
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Language

Interaction Human & Computer

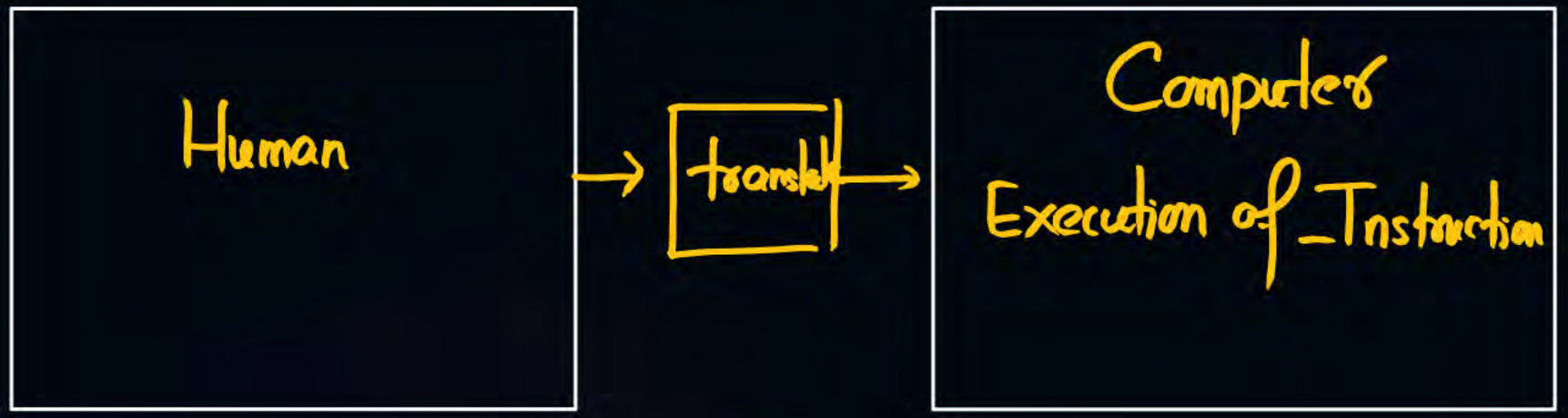
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Computer and its Language



Slide

Basic task of Computer



Execution of program.

program is

Sequence of Instruction

language difference

Slide



Computer Understand the language

Computer understand binary language

0/1 – word, alphabet, number, pic, movie



String of 0's & 1's

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Machine Language

10100001 10111100 10010011 00000100
00001000 00000011 00000101 11000000
10010011 00000100 00001000 10100011
11000000 10010100 00000100 00001000

Language of Computers

Nonsense

It means: z = x + y; ← If carries sense

Slide



Machine Language

10100001 10111100 1001001 00000100
00001000 00000011 00000101 11000000
10010011 00000100 00001000 10100011
11000000 10010100 00000100 00001000

Machine Language

The language

M/c understand

It means: z = x + y;

Slide



Assembly Language

Assembly language uses mnemonic instructions to represent machine code instructions. These are short, easy-to-remember words representing specific instructions that the computer's processor can understand.

Mnemonic Ins

Machin code instruction → Code

01100000 Add

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Topic : Assembly Language: Low Level Language

Low level language is more closer to M/c language

Slide



Topic : Assembly Language: Low Level Language

We call those languages as low level languages which are closer to hardware as compared to high-level languages instead of software.

they allow programmers to manipulate hardware elements like register, memory etc.

Low-level languages are often used for designing systems, such as developing operating systems, device drivers, and embedded systems.

Assembly Language

8085

Mnemonics	Hex code
MVI A, 20H	3EH
	20H
MVI B, 40H	06H
	40H
ADD B	80H
STA 2200H	32H
	00H
	22H
HLT	76H

MUI A 20H

3E H ← Hexadecimal No.

0011 1110

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High level language

High level languages are programming languages which are used for writing programs or software which could be understood by the humans and computer. High level languages are easier to understand for humans because it uses lot of symbols letters phrases to represent logic and instructions in a program. It contains high level of abstraction compared to low level languages.

High level language

1. programming

2. Human understandle

Symbol

Slide

High level language is a language that in human understandable form



High level language

Basically symbolic languages that use English words and/or mathematical symbols rather than mnemonic codes.

English word / Mathematical symbol

C is High level + Low level language



High level language

More closure to human language and further from machine language

Slide



High Level language

C = A + B;

C | C++ | JAVA

High Level Language

translate

ADD A , B — Assembly Language

100100111 — Machine Language

☑ — Hardware



High Level language



An Instruction will execute

Slide



Topic : Translation of one language to another

Text editor program

+

Compiler ← Software / System program

↓

M/C code which is 0101110.... understand by M/C executable code

Slide



Topic : Translation of one language to another

Executable code : group of M/c Instructions which

can run of M/c.

Slide



Topic : Translation of one language to another

Compiler : Language translator

Language is syntax

Syntax checking

Slide



Topic : C is middle Level language

C is considered a middle-level language because it has features of both low-level and high-level languages. For example, C supports pointer arithmetic, which is low-level, but it is also machine independent, which is high-level. C also gives programmers access to direct manipulation with the computer hardware, which higher-level languages do not allow



Topic : C Developed By

Dennis Ritchie. 1972. Bell Laboratories



Slide



Topic : What is a Program?

A program is a piece of code or set of instructions that tells a computer how to perform a task.



Topic : Application of C

1. System Programming:

Operating Systems:

C is used to build the core of many operating systems like Windows, Linux, and macOS. Its low-level access to memory and hardware makes it ideal for writing efficient and powerful system software.



Topic : Application of C

Device Drivers:

These are software components that allow the operating system to interact with hardware devices. C's ability to directly manipulate hardware makes it a natural choice for this purpose.



Topic : Application of C

Compilers and Assemblers:

Compilers and assemblers translate high-level programming languages into machine code. C is often used to write these tools due to its efficiency and ability to generate optimized code.



Topic : Application of C

2. Embedded Systems:

Microcontrollers:

C is widely used to program microcontrollers found in devices like washing machines, microwaves, and cars. Its small footprint and efficiency make it suitable for resource-constrained environments.



Topic : Application of C

IoT Devices:

The Internet of Things (IoT) relies on small, connected devices. C's performance and ability to interact with hardware make it a popular choice for developing IoT applications.



Topic : Application of C

3. Application Software:

Text Editors and Compilers:

Some of the most popular text editors and compilers, like Vim and GCC, are written in C.



Topic : Application of C

Databases:

C is used in the development of database management systems like MySQL and PostgreSQL. It allows for efficient data storage and retrieval.



Topic : Application of C

Games:

While not as common as C++ for game development, C is still used in performance-critical parts of game engines and for developing games with smaller scopes.



Topic : Application of C

4. Networking:

Network Protocols:

C is used to implement network protocols like TCP/IP, which are essential for communication between devices on a network.



Topic : Application of C

Web Servers:

High-performance web servers like Apache and Nginx are partially written in C for efficiency and low-level control.



Topic : Application of C

5. Other Applications:

Financial Applications:

High-frequency trading systems and other financial applications that require fast execution speeds often use C.

Scientific and Engineering Applications:

C is used in scientific simulations, data analysis tools, and engineering software where performance and precision are crucial.



2 mins Summary

Topic — Language

Topic — Low Level Language

Topic — High level language

Topic — Translator

Topic

Slide

THANK - YOU

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